BOUNDED BELOW COMPLEXES ADMIT CARTAN-EILENBERG RESOLUTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7350-7391, following the statement at 7343-7348.

Editorial note. The assembly check is 'everything is clear' (7390) after every ingredient is displayed; the defining conditions of a Cartan-Eilenberg resolution are read off the compatible short exact sequences, recorded.

Allowed standard prerequisites (editorial).

- Injective resolutions exist. lemma-injective-resolutions-exist, cited at proof lines 7368 and 7361 context; complete proof at source lines 6511-6522, queued for separate carding. Supplies the initial injective resolution of Z^n and the boundedness control of resolutions.
- Compatible resolutions of short exact sequences. lemma-injective-resolutionses, cited at proof lines 7369 and 7376; its source proof at 6821-6880 is complete (the lemma's statement carries an author placeholder item (4) with no content). Supplies the resolution triples (I^{n,q}, J_B^{n+1,q}) and (J_Z^{n+1,q}, J_H^{n+1,q}) with compatible exact sequences.

Source reference. Original lemma 013T: sources/stacks-derived/pinned-original-source.tex, source lines 6788–6788. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 013K: sources/stacks-derived/pinned-original-source.tex, source lines 6497–6497. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category with enough injectives. Let $K^\bullet$ be a bounded below complex. There exists a Cartan-Eilenberg resolution of $K^\bullet$.*

Proof. Suppose that $K^p = 0$ for $p < n$. Decompose $K^\bullet$ into short exact sequences as follows: Set $Z^p = \mathrm{Ker}(d^p)$, $B^p = \mathrm{Im}(d^{p-1})$, $H^p = Z^p/B^p$, and consider

$$\begin{aligned} 0 \to Z^n \to K^n \to B^{n+1} \to 0 \\ 0 \to B^{n+1} \to Z^{n+1} \to H^{n+1} \to 0 \\ 0 \to Z^{n+1} \to K^{n+1} \to B^{n+2} \to 0 \\ 0 \to B^{n+2} \to Z^{n+2} \to H^{n+2} \to 0 \\ \ldots \end{aligned}$$

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Set $I^{p,q} = 0$ for $p < n$. Inductively we choose injective resolutions as follows:

(1) Choose an injective resolution $Z^n \to J_Z^{n,\bullet}$ with $J_Z^{n,m} = 0$ for $m < 0$, see Lemma 013K.
(2) Using Lemma 013T choose injective resolutions $K^n \to I^{n,\bullet}$, $B^{n+1} \to J_B^{n+1,\bullet}$ with $I^{n,m} = 0$ for $m < 0$ and $J_B^{n+1,m} = 0$ for $m < 0$, and an exact sequence of complexes $0 \to J_Z^{n,\bullet} \to I^{n,\bullet} \to J_B^{n+1,\bullet} \to 0$ compatible with the short exact sequence $0 \to Z^n \to K^n \to B^{n+1} \to 0$.
(3) Using Lemma 013T choose injective resolutions $Z^{n+1} \to J_Z^{n+1,\bullet}$, $H^{n+1} \to J_H^{n+1,\bullet}$ with $J_Z^{n+1,m} = 0$ for $m < 0$ and $J_H^{n+1,m} = 0$ for $m < 0$ and an exact sequence of complexes $0 \to J_B^{n+1,\bullet} \to J_Z^{n+1,\bullet} \to J_H^{n+1,\bullet} \to 0$ compatible with the short exact sequence $0 \to B^{n+1} \to Z^{n+1} \to H^{n+1} \to 0$.
(4) Etc.

Taking as maps $d_1^\bullet : I^{p,\bullet} \to I^{p+1,\bullet}$ the compositions $I^{p,\bullet} \to J_B^{p+1,\bullet} \to J_Z^{p+1,\bullet} \to I^{p+1,\bullet}$ everything is clear. $\square$